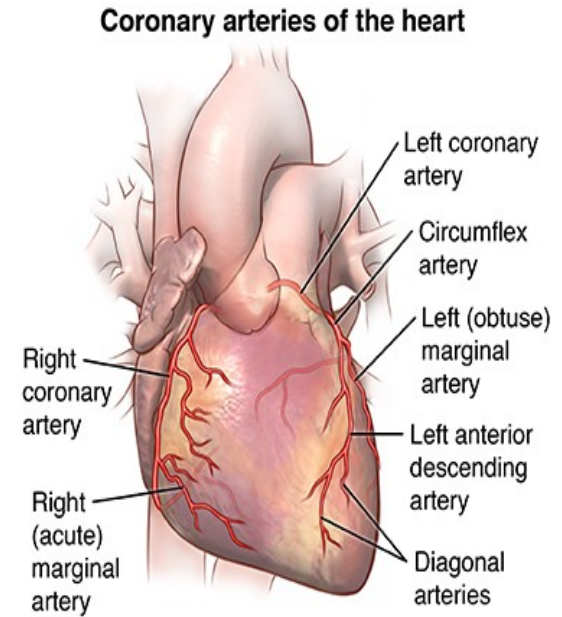
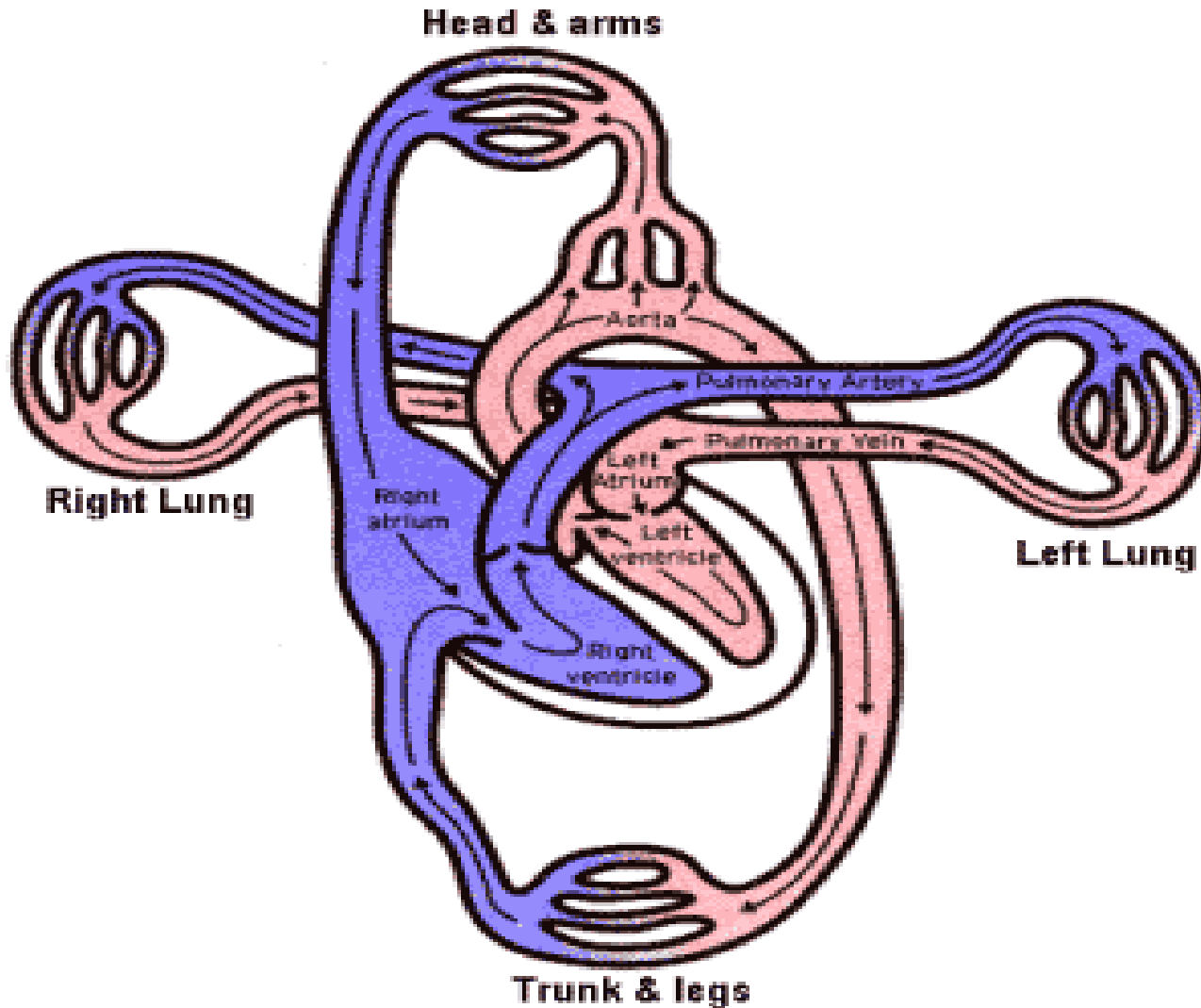


Pharmacotherapy of Heart Failure

- # What is Heart Failure?



Introduction

- ▶ **Heart failure (HF)**

- ▶ a clinical syndrome caused by the inability of the heart to pump sufficient blood to meet the metabolic demands of the body

- ▶ Result from any abnormality in cardiac structure or function that reduces:

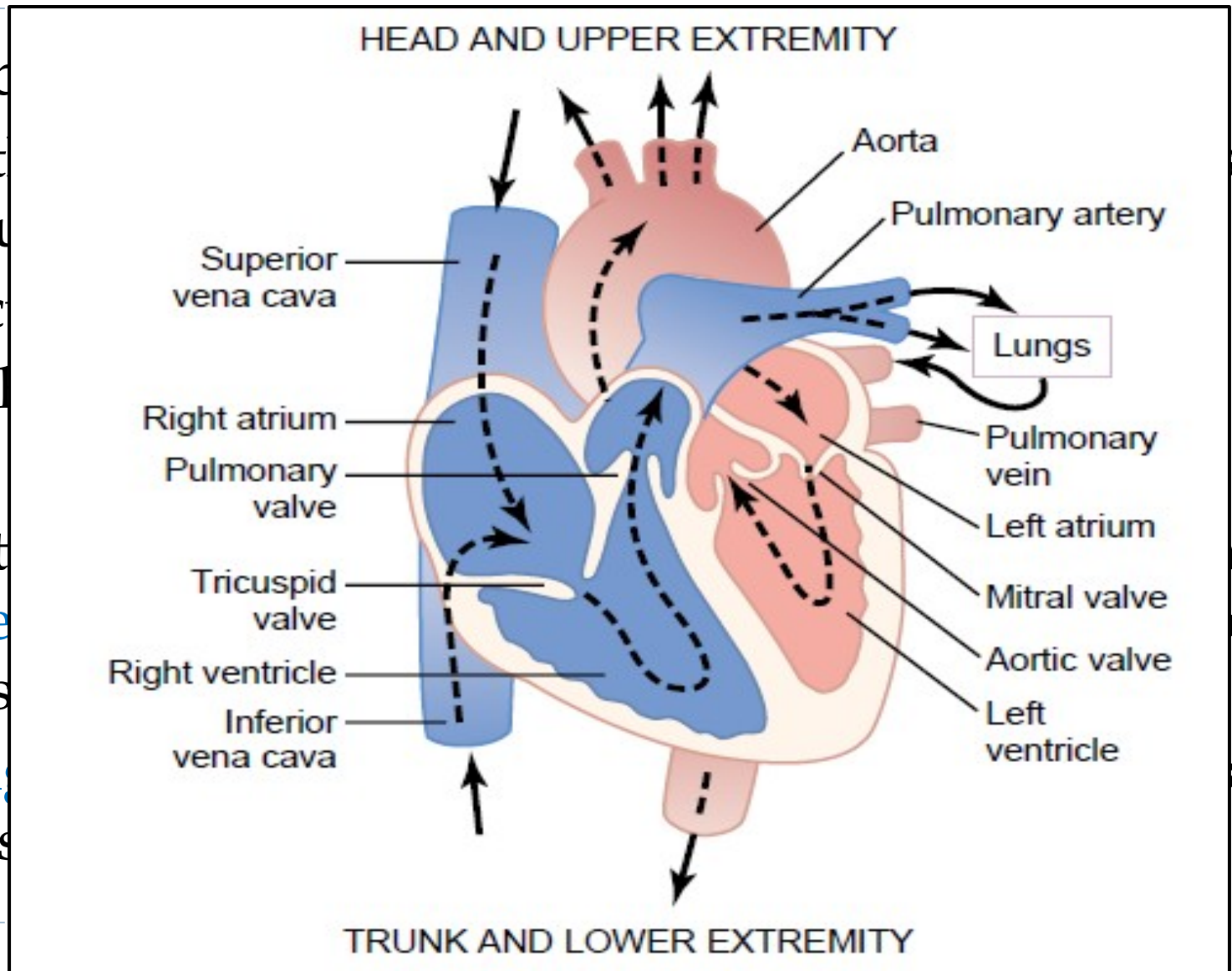
- ▶ ventricular filling (diastolic dysfunction) and/or

- ▶ myocardial contractility (systolic dysfunction)



Introduction

- ▶ HF can be caused by:
 - ▶ When there is a problem in the lungs
 - ▶ Difficult to breathe
 - ▶ Breathing becomes labored (SOB)
 - ▶ When there is a problem with the **feet** (peripheral edema) progresses to **Pitting** edema, which leaves



Epidemiology

- ▶ HF is a major medical problem, with a substantial economic impact.
- ▶ Affects more than 64 million People globally
 - 1000,000 new cases each year
- ▶ A large majority of patients with HF are elderly,
 - ▶ with multiple comorbidity that influence morbidity and mortality
 - ▶ the leading cause of hospitalization in elderly patients
- ▶ The overall 5-year survival ~42% for all patients with a diagnosis of HF



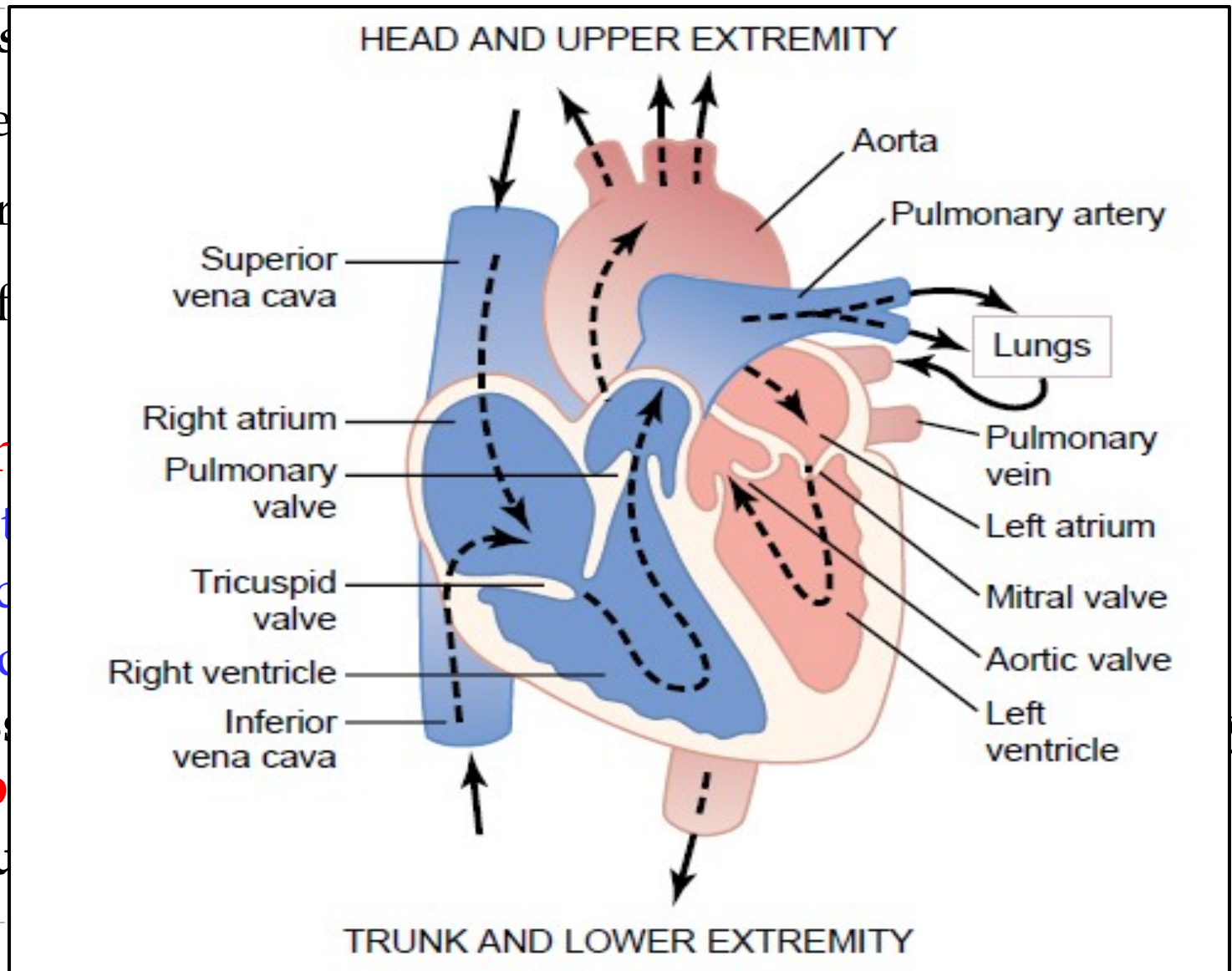
Etiology

▶ *Systolic dys*

- ▶ When the
- ▶ HF with r
- ▶ Ejection f

Causes:

- ▶ **Coron**
- ▶ **Reduct**
- ▶ **Dilated**
- ▶ **Ventric**
- **Press**
- hyp**
- **Volu**



Etiology

- ▶ *Diastolic dysfunction (restriction in ventricular filling)*
 - ▶ HF with preserved ejection fraction (HFpEF)...50% of pts with HF
 - ▶ EF is normal or slightly abnormal

Causes:

- ▶ Increased **ventricular stiffness** disturbances in relaxation of the heart
 - **Ventricular hypertrophy**
 - **Infiltrative myocardial diseases** (e.g., amyloidosis, endomyocardial fibrosis)
 - **Myocardial ischemia and infarction**
 - ▶ Mitral or tricuspid valve stenosis
 - ▶ Pericardial disease (e.g. pericarditis and pericardial tamponade)
-



Pathophysiology

- ▶ When heart fails, CO is reduced

$$CO = HR \times SV$$

- Heart rate

- Controlled by the ANS

- Stroke volume: depends on

- Preload
 - Afterload and
 - Contractility



Pathophysiology

➤ Preload

- The ability of the heart to alter the force of contraction depends on changes in preload
 - According to Frank-Starling mechanism
 - $\uparrow$ in preload----- $\uparrow$ myocardial sarcomere length----- $\uparrow$ number of cross-bridges between thick (myosin) and thin (actin) myofilaments----- $\uparrow$ force of contraction----- $\uparrow$ SV----- $\uparrow$ CO
 - The length of the sarcomere is determined primarily by the volume of blood in the ventricle.
 - Therefore, left ventricular end-diastolic volume (LVEDV) is the primary determinant of preload.
-



Pathophysiology

▶ **Afterload**

- ▶ the sum of forces preventing active forward ejection of blood by the ventricle
- ▶ In patients with left ventricular systolic dysfunction, increasing afterload causes a decrease in SV

▶ **Contractility**

- ▶ the intrinsic property of cardiac muscle
- ▶ describes fiber shortening and tension development



Pathophysiology

- ▶ As a result of decrease in CO in heart failure
- ▶ the heart will rely on compensatory responses to maintain an adequate CO
- ▶ These compensatory responses include
 - ▶ Activation of SNS
 - ▶ Activation of RAAS
 - ▶ Increased preload
 - ▶ Vasoconstriction
 - ▶ **ventricular hypertrophy and remodeling**
 - ▶ changes in ventricular size, shape, structure, and function



Pathophysiology

- ▶ Compensatory responses
 - ▶ intended to provide short-term support to maintain circulatory homeostasis
 - ▶ However, long-term activation
 - ▶ results in complex functional, structural, biochemical, and molecular changes that leads to heart failure



Common HF precipitating factors

Conditions

- Noncompliance with therapy,
- Coronary ischemia,
- Inappropriate medication use,
- Cardiac events (e.g., MI, AF),
- Different endocrine disorders
- Pulmonary **infections**, and
- **Anemia.**

Drugs may precipitate HF

- Cardiotoxic drugs
 - (e.g. Doxorubicin, cyclophosphamide...)
- Sodium and Water retention
 - ✓ NSAIDS, Glucocorticoids,
 - ✓ Na⁺ containing drugs (**diclofenac sodium**)
- Negative Inotropic effects
 - B-Blockers, CCBs
 - Antiarrhythmic drugs



Clinical Presentations

▶ **General**

- ▶ Patient presentation may range from **asymptomatic to cardiogenic shock**

▶ **Symptoms**

- ▶ **Dyspnea, particularly on exertion**
- ▶ **Orthopnea**
- ▶ **Paroxysmal nocturnal dyspnea (PND)**
- ▶ **Exercise intolerance**
- ▶ **Tachypnea**

▶ **Cough**

▶ **Fatigue**

▶ **Nocturia**

▶ **Hemoptysis**

▶ **Abdominal pain**

▶ **Anorexia**

▶ **Nausea**

▶ **Bloating**

▶ **Fluid overload/Ascites**

▶ **Mental status changes**



Signs and symptoms

▶ Signs

- ▶ Pulmonary rales
- ▶ Pulmonary edema
- ▶ **S₃ gallop**
- ▶ Cool extremities
- ▶ Pleural effusion
- ▶ Tachycardia
- ▶ Narrow pulse pressure
- ▶ Cardiomegaly
- ▶ **Peripheral edema**
- ▶ JVD (may be >4cm)
- ▶ Hepatojugular reflux
- ▶ Hepatomegaly
- ▶ Cheyne–Stokes respiration



Signs and symptoms

▶ Laboratory Tests

- ▶ **BNP >100 pg/mL**
- ▶ Electrocardiogram
 - may be normal, or acute ST-T wave changes
- ▶ Elevated Serum creatinine
- ▶ Complete blood count (CBC) (Anemia)
- ▶ Chest x-ray: useful for detecting
 - cardiac enlargement
 - pulmonary edema, and
 - pleural effusions



Signs and symptoms

- ▶ **Echocardiogram**: used to assess
 - the **size** of the left ventricle,
 - valve function
 - pericardial effusion
 - wall motion abnormalities, and
 - **ejection fraction reduced or normal**
- ▶ Hyponatremia: serum sodium <130 mEq/L is associated with reduced survival and may indicate worsening volume overload and/or disease progression



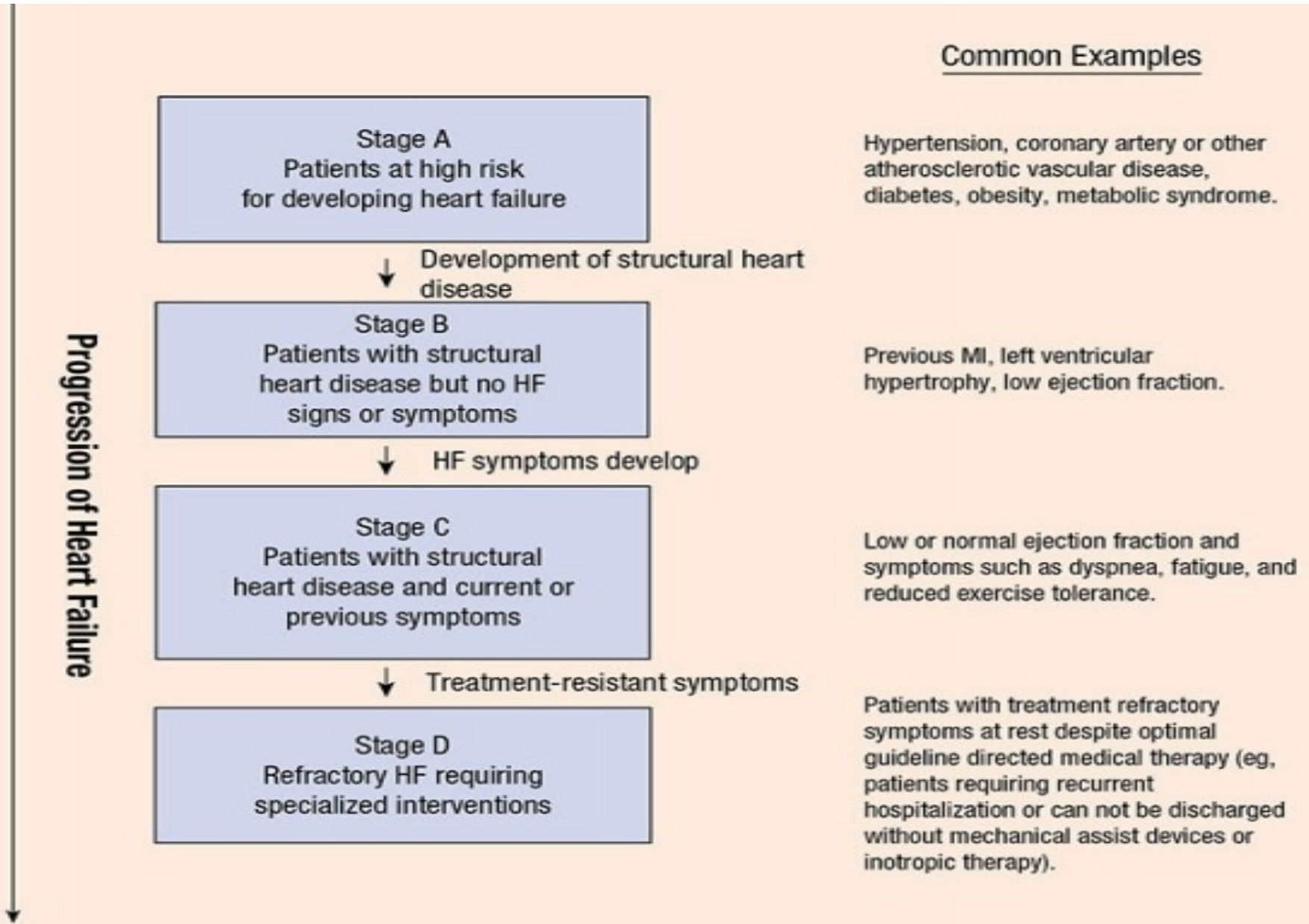
Diagnosis

- ▶ No single test to confirm the diagnosis
 - ▶ A complete history and physical examination targeted at identifying cardiac or noncardiac disorders
 - ▶ A careful medication history
 - ▶ Laboratory testing
 - ▶ BNP
 - ▶ CBC
 - ▶ hemoglobin A1C
 - ▶ electrolytes (including calcium and magnesium)
 - ▶ thyroid function tests
 - ▶ chest x-ray
 - ▶ ECG
-



Classification

ACC/AHA classification



Classification

NYHA classification

NYHA class

- | | |
|-----|---|
| I | Patients with cardiac disease but without limitations of physical activity . Ordinary physical activity does not cause undue fatigue, dyspnea, or palpitation |
| II | Patients with cardiac disease that results in slight limitations of physical activity . Ordinary physical activity results in fatigue, palpitation, dyspnea, or angina. |
| III | Patients with cardiac disease that results in marked limitation of physical activity . Although patients are comfortable at rest, less than ordinary activity will lead to symptoms |
| IV | Patients with cardiac disease that results in an inability to carry on physical activity without discomfort . Symptoms of congestive heart failure are present even at rest . With any physical activity, increased discomfort is experienced. |
-



ACC/AHA Staging and NYHA Classification of HF

<i>ACC-AHA Stage</i>	<i>NYHA Functional Class</i>
A. At high risk for heart failure, but without structural heart disease (normal heart exam) or symptoms of heart failure (e.g., patients with hypertension, coronary heart disease, diabetes, alcoholism, or strong family history)	No corresponding category
B. Structural heart disease present (e.g., LV hypertrophy, dilation, fibrosis, old MI), but without symptoms of heart failure	I. Asymptomatic or no limitations on normal physical activity, but symptoms with strenuous exercise.
C. Structural heart disease with prior or current symptoms of heart failure	II. Symptoms of heart failure with normal activity or with moderate exertion III. Symptomatic with minimal exertion; marked limitations in physical activity including activities of daily life (e.g., bathing, dressing) (or IIIB). Symptoms of heart failure at rest.
D. Refractory heart failure requiring specialized interventions	IV (or IVB). Symptoms of heart failure at rest and requiring hospitalization or intravenous inotropic support

Treatment of Heart failure

Goal of therapy

- ▶ To improve the patient's quality of life
- ▶ To relieve or reduce symptoms
- ▶ To prevent or minimize hospitalizations for exacerbations of heart failure
- ▶ To slow progression of the disease process
- ▶ To prolong survival



Non-pharmacologic Therapy

- ▶ Implantable cardioverter-defibrillator (ICD)
- ▶ Cardiac resynchronization therapy (CRT)
 - In patient with NYHA class III to IV, receiving optimal medical therapy and with a QRS duration ≥ 120 milliseconds and LVEF $\leq 35\%$
- ▶ Restriction of physical activity
 - ▶ For patients with acute congestive symptoms
- ▶ Encourage physical activity
 - ▶ once the patient's symptoms have stabilized and excess fluid is removed
- ▶ restriction of dietary sodium and fluid intake
 - ▶ Mild (<3 g per day)
 - ▶ moderate (<2 g per day) sodium restriction

Non-pharmacologic treatment

- ▶ Restriction of fluid intake → 2 L per day from all sources
 - ▶ In patients with hyponatremia (serum Na <130 mEq/L) or
 - ▶ In those with persistent volume retention despite high diuretic doses and sodium restriction,
- ▶ Immunization against influenza and pneumococcus
 - ▶ Reduce the risk of respiratory infections
- ▶ Avoidance of medications that can exacerbate heart failure



Treatment of Heart Failure

- ▶ **Treatment of Stage A Heart Failure**
 - ▶ identification and modification of risk factors
 - ▶ Treat according to current guidelines
 - ▶ Hypertension
 - ▶ Dyslipidemia
 - ▶ Diabetes
 - ▶ coronary artery disease
 - ▶ metabolic syndrome
 - ▶ Additional risk factors that require modifications includes
 - ▶ Obesity
 - ▶ Smoking



Treatment of Heart Failure

▶ Treatment of Stage B Heart Failure

- ▶ Treatment is targeting at minimizing additional injury and
- ▶ Preventing or slowing the remodeling process
- ▶ In addition to the treatment measures outlined in stage A
 - ▶ ACE inhibitors and beta-blockers are important components of therapy
 - ▶ Patients with a previous MI, reduced LVEF (<40%)
 - ▶ **ACEI+ beta-blockers**
 - ▶ Alternative: ARBs in patients intolerant to ACEI

Treatment

Treatment of Stage C Heart Failure

- ▶ Continue the treatments in Stages A and B
- ▶ In stage C, HF patients should be treated with **three medications**
 - An ACE inhibitor and
 - A β -blocker
 - A diuretic (**if there is evidence of fluid retention**)



Treatment

Treatment of Stage C Heart Failure

- ▶ An aldosterone receptor antagonist should also be considered in selected patients (e.g. Spironolactone)
- ▶ Digoxin therapy
 - For symptom reduction
 - To decrease hospitalizations, or
 - Slow ventricular response in patients with concomitant atrial fibrillation



Treatment of Stage C Heart Failure

Diuretic therapy

- ▶ Recommended in all patients with clinical evidence of fluid retention (Sodium and water retention) that
 - result from activation of the compensatory mechanisms
 - ▶ Produces symptomatic benefits
 - ▶ Improve exercise tolerance and quality of life, and
 - ▶ Reduce hospitalizations from HF
 - ▶ Do not prolong survival or alter disease progression (with the possible exception of torsemide)
 - ▶ **Loop diuretics are the most potent.**
-



Treatment of Stage C Heart Failure

- *Loop Diuretics includes*

- **Furosemide:** 20-160 mg QD or BID

- CrCl 20-50 ml/min: 160 mg QD or BID

- CrCl <20 ml/min: 400 mg QD

- **Bumetanide:** 0.5-2 mg QD or BID

- CrCl 20-50 ml/min: 2 mg QD or BID

- CrCl <20 ml/min: 8-10 mg QD

- **Torsemide:** 10-80 mg QD

- CrCl 20-50 ml/min: 40 mg QD

- CrCl <20 ml/min: 200 mg QD

- ▶ Monitor daily morning body weights, volume depletion, serum electrolytes, SCr level

- Gain of 1.4 to 2.3 kg in a week: increase dose of diuretic



Treatment of Stage C Heart Failure

ACE Inhibitors

- ▶ Decrease the production of angiotensin II and in turn aldosterone
 - ▶ Reduce HF progression including
 - Ventricular remodeling
 - Myocardial fibrosis
 - Myocyte apoptosis
 - Cardiac hypertrophy
 - NE release
 - Vasoconstriction and
 - Sodium and water retention
-



Treatment of Stage C Heart Failure

ACE Inhibitors

- ▶ Improve survival by 20% to 30%
- ▶ Reduce the risk of death or hospitalization
- ▶ Slow the progression of HF, and
- ▶ Reduce the rate of reinfarction (Post-MI)



Treatment of Stage C Heart Failure

ACE Inhibitors

► Includes

- Captopril: 6.25-50 mg TID
 - Enalapril: 2.5-20 mg BID
 - Lisinopril: 2.5- 40 gm QD
 - Fosinopril: 5-40 mg QD
 - Ramipril: 1.25 mg-5 mg BID
-
- Start with low dose and titrate to tolerable dose



Treatment of Stage C Heart Failure

β -Blockersantagonize the detrimental effects of the SNS

- ▶ should be used **in all stable patients** with HFrEF
 - In the absence of contraindications or intolerance
 - Can be initiated before optimizing ACE inhibitor doses
 - Recommended for **asymptomatic patients with a reduced left ventricular EF (Stage B)** to decrease the risk of progression to HF



Treatment of Stage C Heart Failure

▶ β -Blockers

▶ Reduce morbidity and mortality

- Decrease ventricular mass

- Improve the sphericity of the ventricle

- Reduce systolic and diastolic volumes

*Reverse
remodeling*

- ▶ Decrease myocyte death from catecholamine-induced necrosis or apoptosis

- ▶ Decrease HR and ventricular wall stress

- ▶ Inhibit plasma renin release



Treatment of Stage C Heart Failure

β -Blockers

- ▶ Three β -blockers have been shown to significantly reduce mortality in Randomized Control Trial
 - Carvedilol: 10-80 mg QD
 - Metoprolol succinate (CR/XL): 12.5-200 mg QD
 - Bisoprolol: 1.25-10 mg QD



Treatment of Stage C Heart Failure

β -Blockers

- ▶ Should be initiated in stable patients who have no or minimal evidence of fluid overload
 - ▶ Greater magnitude of HR reduction was significantly associated with greater improvement in survival.
 - ▶ To minimize acute decompensation
 - Start at a very low doses with slow upward dose titration
 - Titrate dose at no more often than every 2 weeks
-



Treatment

▶ Treatment of Stage C Heart Failure

▶ According to the 2016 ACC/AHA/HFSA Focused Update on New Pharmacological Therapy for Heart Failure

▶ For patients with stage C HF

▶ inhibition of the RAAS with

- ACE inhibitors or

- ARBs or

- Angiotensin receptor neprilysin inhibitor (ARNI)

- B-blockers and aldosterone antagonists is recommended



Treatment

- ▶ In patients with chronic symptomatic HFrEF NYHA class II or III who tolerate an ACE inhibitor or ARB
 - replacement by an ARNI is recommended to further reduce morbidity and mortality
- ▶ In ARNI
 - ▶ an ARB is combined with an inhibitor of neprilysin
 - an enzyme that degrades natriuretic peptides, bradykinin, adrenomedullin, and other vasoactive peptides
 - Includes
 - **valsartan/sacubitril**: 160/40 mg BID
 - Side effect of ARNI includes
 - Hypotension, renal insufficiency and angioedema

Treatment

- ▶ ARNI should not be administered concomitantly with ACE inhibitors or within 36 hours of the last dose of an ACE inhibitor
- ▶ ARNI should not be administered to patients with a history of angioedema



Treatment

Treatment of Stage D Heart Failure

- ▶ Includes patients
 - With refractory symptoms at rest despite maximal medical therapy and
 - Who undergo recurrent hospitalizations or cannot be discharged from the hospital without special interventions
- In addition to standard treatments outlined in Stages A to C
 - ▶ Specialized therapies
 - mechanical circulatory support
 - continuous IV positive inotropic therapy, and
 - cardiac transplantation



Treatment

Treatment of Stage D Heart Failure

- ▶ Less tolerant to ACE inhibitors (hypotension, worsening renal insufficiency) and β - blockers (worsening HF)
 - Use low doses, slow upward dose titration, and close monitoring for signs and symptoms
- ▶ High doses of diuretics.....aggressive diuresis
- ▶ Combination therapy with a loop and thiazide diuretic
- ▶ Restriction of sodium and fluid intake



Drug Therapies in Selected Patients

Angiotensin II Receptor Blockers

- ▶ Used in patients intolerant to ACEIs (cough)
 - ▶ Includes
 - Valsartan: 20-160 mg BID
 - Candesartan: 4-32 mg QD
 - Losartan: 25-150 mg QD
 - ▶ Should not be used in combination with ACEIs (hyperkalemia)
 - ▶ Use with caution in patients with a history of ACEIs associated angioedema
-



Drug Therapies in Selected Patients

Aldosterone Antagonists

- ▶ Aldosterone antagonists attenuate
 - Cardiac extracellular matrix and collagen deposition
 - Cardiac fibrosis and ventricular remodeling
 - The systemic proinflammatory state, atherogenesis, and oxidative stress caused by aldosterone
 - Aldosterone-induced calcium excretion and reductions in bone mineral density
-



Drug Therapies in Selected Patients

Aldosterone Antagonists

- ▶ Adding a low-dose aldosterone antagonist to standard therapy
 - Improve symptoms
 - Reduce the risk of HF hospitalization, and
 - Increase survival
 - ▶ low-dose aldosterone antagonists are appropriate for patients
 - With mild to moderately severe systolic HF (NYHA class II to IV)
 - Post MI with left ventricular dysfunction and either acute HF or diabetes
-



Drug Therapies in Selected Patients

Aldosterone Antagonists includes:

- Spironolactone: 12.5-50 mg QD
 - Eplerenone: 25-50 mg QD
 - ▶ Hyperkalemia is common in HF patients receiving aldosterone antagonists
 - ▶ **To reduce incidence of hyperkalemia**
 - ▶ Avoid aldosterone antagonists in patients with
 - $\text{Scr} > 2 \text{ mg/dl}$ in women or $> 2.5 \text{ mg/dl}$ in men or a $\text{CrCl} < 30 \text{ ml/min}$
 - Recent worsening of renal function
 - Serum potassium concentration $\geq 5 \text{ meq/l}$
 - History of severe hyperkalemia
 - ▶ Start with low dose
-

Drug Therapies in Selected Patients

To reduce incidence of hyperkalemia

- ▶ Avoid concomitant use of NSAIDs or COX-2 inhibitors
 - ▶ Avoid concomitant use of high dose ACEIs, and ARBs with aldosterone antagonists
 - ▶ Avoid triple therapy with ACEI, ARB and aldosterone antagonists
 - ▶ Monitor serum K⁺ conc. and renal function 3 days and 1 wk after initiation and then monthly for the first three months and every three months thereafter
 - ▶ If serum K⁺ conc. > 5.5 at any point during therapy, discontinue any K⁺ supplement or reduce dose of aldosterone antagonist therapy
-



Drug Therapies in Selected Patients

Digoxin

- MOA: exerts its **positive inotropic** effect by inhibiting Na⁺K⁺ATPase which facilitates Ca²⁺ entry into the cell
 - attenuates the excessive SNS activation present in HF patients
 - Cause an increase in parasympathetic activity: ↓ HR
 - Enhancing diastolic filling
 - Result in slowed conduction and prolongation of AV node refractoriness
 - Slow the ventricular response in patients with AF
 - ▶ Benefits in HF are related to its neurohormonal modulating activity
-



Drug Therapies in Selected Patients

Digoxin

- ▶ digoxin improves
 - cardiac function
 - quality of life
 - exercise tolerance, and
 - HF symptoms in patients with **Systolic Heart Failure** (EF<40%)
- ▶ **Unknown effect on mortality**



Drug Therapies in Selected Patients

Digoxin

- ▶ Place in therapy of HF
 - ▶ **Early:** in patients with HF and supraventricular tachyarrhythmias such as atrial fibrillation
 - To control ventricular response rate
 - ▶ To reduce symptoms: In patients with normal sinus rhythm
 - should be used in conjunction with other standard HF therapies including diuretics, ACE inhibitors, and β -blockers



Drug Therapies in Selected Patients

Digoxin

- ▶ Dose: 0.125-0.25 mg QD
 - ▶ Target plasma concentration: 0.5-1 ng/ml (0.6 to 1.3 nmol/L)
- ▶ **Reduce dose:**
 - Decreased renal function, the elderly, or those receiving interacting drugs (e.g., amiodarone)
 - Should receive 0.125 mg daily or every other day

Digoxin withdrawal results in

- Worsening of HF
 - Decreased exercise capacity, and
 - Reduction in ejection fraction
-



Drug Therapies in Selected Patients

Nitrates and Hydralazine

- ▶ Complementary hemodynamic actions
 - Isosorbide dinitrate: Venodilation, decrease preload
 - Hydralazine: arterial vasodilator, decrease SVR and increases SV and CO
- ▶ Dose:
 - Hydralazine: 37.5-75 mg TID
 - ISDN: 20-40 mg TID



Drug Therapies in Selected Patients

Nitrates and Hydralazine

- ▶ Particularly effective in African Americans
 - ▶ **Added to the standard three drug regimen**
- ▶ **Recommended for patients unable to tolerate either an ACE inhibitor or ARB**
- ▶ **Reduce mortality**, hospitalizations for HF and improve quality of life



Treatment of Concomitant Disorders

HypertensionTarget BP < 130/80

- ▶ Treat both disorders by ACEIs, β -blockers, and diuretics
 - ▶ If control of HTN is not achieved add
 - aldosterone antagonist, ISDN/hydralazine, or amlodipine (or possibly felodipine)
 - ▶ **Avoid the following medications** in patients with Systolic HF(HFrEF)
 - ▶ verapamil, diltiazem and direct acting vasodilators (e.g., minoxidil)
 - sodium retention
 - ▶ In patients with HFpEF, both verapamil and diltiazem can be safely used
-



Treatment of Concomitant Disorders

Angina

- ▶ Nitrates and β -blockers are effective antianginals and are the preferred agents for patients with both disorders
 - Improve hemodynamics and clinical outcomes
- ▶ Control fluid retention with diuretics
- ▶ Alternatives: amlodipine and felodipine



Treatment of Concomitant Disorders

Atrial Fibrillation

- **Digoxin:** used to slow ventricular response in patients with **HF and atrial fibrillation**
- β -Blockers are more effective than digoxin
 - Improve morbidity and mortality in patients with SHF
- Combination therapy with digoxin and a β -blocker: more effective
 - Alternative; Amiodarone
- Non-dihydropyridine CCBs such as verapamil or diltiazem should be avoided
- Antithrombotic therapy for stroke prevention



Treatment of Concomitant Disorders

Diabetes

- ▶ **Avoid the TZDs** (pioglitazone and rosiglitazone) in patients with **diabetes and HF**
 - **Fluid retention**
- ▶ Absolute contraindication: NYHA class III or IV HF
- ▶ Relative contraindication: NYHA class I or II



Treatment of Heart failure with preserved ejection fraction(HEpEF)

- ▶ With a few notable exceptions, many of the drugs used to treat SHF are the same as those for treatment of HFpEF
- ▶ Difference
 - The rationale for their use
 - The pathophysiologic process that is being altered by the drug, and
 - The dosing regimen



Treatment of HEpEF

- ▶ For example
 - ▶ β -blockers
 - ▶ In HFpEF, used to
 - decrease HR
 - increase diastolic duration, and
 - modify the hemodynamic response to exercise
 - ▶ **Diuretics**
 - The doses in HFpEF smaller than those used to treat SHF
- ▶ **Antagonists of the RAAS:** useful to
 - Lower BP and
 - reducing Left Ventricular Hypertrophy



Treatment of HEpEF

- ▶ Calcium channel blockers
- ▶ Diltiazem and verapamil **limited use in the treatment of Systolic HF but useful in the treatment of HFpEF**
 - ▶ decrease HR and increase exercise tolerance
- ▶ Important for patients with HTN and coronary artery disease
- ▶ Nondihydropyridines:
 - Verapamil: 20 to 240 mg/day
 - Diltiazem: 90 to 120 mg/day
- ▶ Dihydropyridines:
 - Amlodipine: 2.5 mg/day



TREATMENT OF ACUTE DECOMPENSATED HF

GENERAL APPROACH

- ▶ The term *decompensated HF* refers to pts with **new or worsening signs or symptoms** that are usually caused by **volume overload and/or hypoperfusion** and lead to the need for **additional medical care**, such as emergency department visits and hospitalizations.

-
- ▶ **ECG monitoring**, continuous **pulse oximetry**, urine flow monitoring, and **automated BP** recording are necessary.
 - ▶ Precipitating factors of decompensation should be addressed and corrected.
 - ▶ Drugs that may aggravate HF should be discontinued if possible.
 - ▶ **Aggressive diuresis** with IV loop diuretics

Positive Inotropic Agents

- ▶ **Dobutamine** is β_1 - and β_2 -receptor agonist with some α_1 -agonist effects.
 - ▶ The net vascular effect is usually **vasodilation**.
- ▶ It has a **potent inotropic effect** without producing a significant **change in heart rate**.
- ▶ Initial doses of 2.5 to 5 mcg/kg/min can be increased progressively to 20 mcg/kg/min on the basis of clinical and hemodynamic responses.

Milrinone: A phosphodiesterase inhibitor, produces positive **inotropic** and arterial and venous **vasodilating effects**; hence, milrinone has been referred to as an **inodilator**.

- ▶ During IV administration, milrinone increases **stroke volume (CO)** with **little change in heart rate**.

Dopamine: should generally be **avoided in decompensated HF**, but preferred in pts with **marked systemic hypotension or cardiogenic shock**.

Vasodilators

- ▶ **Arterial vasodilators** act as impedance-reducing agents, reducing afterload and causing a reflex increase in CO.
- ▶ **Venodilators** act as preload reducers by increasing venous capacitance, reducing symptoms of pulmonary congestion in pts with high cardiac filling pressures.
- ▶ **Mixed vasodilators** act on both arterial resistance and venous capacitance vessels, reducing congestive symptoms while increasing CO.
 - Sodium nitroprusside
 - *Nitroglycerin*
 - *Nesiritide*

MECHANICAL CIRCULATORY SUPPORT

- ▶ Intra-aortic Balloon Pump
- ▶ Ventricular Assist Devices

SURGICAL THERAPY

- ▶ cardiac transplantation

EVALUATION OF THERAPEUTIC OUTCOMES

CHRONIC HEART FAILURE

- ▶ Presence and severity of symptoms , fluid retention
- ▶ Improvement in exercise tolerance and fatigue, decreased nocturia, and a decrease in heart rate.
- ▶ Routine BP, body weight, serum electrolytes and renal function monitoring is necessary.
- ▶ Adverse events of drugs the patient is taking

ACUTE DECOMPENSATED HF

- ▶ Initial stabilization requires achievement of adequate **arterial oxygen saturation** and content
- ▶ Cardiac index and BP must be sufficient to ensure adequate organ perfusion
 - ▶ Goals of cardiac index ($>2.2\text{L/min/m}^2$) and BP (MAP $> 60\text{mm Hg}$)